

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow blood bank staff to record and update blood inventory based on blood group and available units.	High	When staff add, issue, or update blood units, the corresponding blood group inventory count is updated correctly and displayed.	Maintains accurate information about available blood stock.
FR-002	Functional	The system shall allow authorized users to check the availability of blood units for a specified blood group.	High	When a blood group is selected, the system displays its current available units.	Helps staff determine whether the required blood type is available.
FR-003	Functional	The system shall allow authorized users to submit an emergency blood request specifying the required blood group and number of units.	High	When valid blood group and unit requirements are entered, the system creates and displays an emergency request with a unique request ID.	Enables urgent blood requirements to be registered and processed.
FR-004	Functional	The system shall identify eligible donors matching the	High	Given a valid emergency request, the system displays registered donors	Enables rapid identification of potential emergency donors.

		required blood group for an emergency blood request.		whose blood group matches the requested blood group and who are marked as available.	
FR-005	Functional	The system shall notify an eligible donor when the donor is selected for an emergency blood request.	High	When an eligible donor is selected, the system sends a notification and records the notification status for that donor.	Allows eligible donors to respond quickly to emergency requirements.
NFR-001	Non-Functional	The system shall display the current blood inventory for a requested blood group within 2 seconds.	High	In at least 9 out of 10 tests, the inventory result is displayed within 2 seconds of the request.	Fast inventory access is important when handling urgent blood requirements.
NFR-002	Non-Functional	The system shall restrict blood inventory and donor information to authorized users.	High	An unauthenticated or unauthorized user attempting to access inventory or donor information is denied access.	Protects sensitive blood bank and donor information.

BLOOD BANK INVENTORY AND EMERGENCY DONOR